ISHAN FERNANDES

Software Engineer · Game Developer

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PROFILE

CS student at Royal Holloway, University of London who ships and not just studies. Built 5 original browser games and full-stack projects from scratch, including a platformer with a working ghost replay system and a probabilistic AI opponent. Writes clean JavaScript and Python, understands software architecture at a design-pattern level, and actively expands into game engines (Godot) and cybersecurity. Looking for an internship where real code ships.

PROJECTS & WORK

One Minute Left: Browser Platformer | ishanfernandes.uk/projects 2026

- Engineered a complete game loop in vanilla JS + Canvas API, delivering sub-16ms frame rendering with no external engine dependency.
- Designed and implemented a ghost replay system from scratch — recording and replaying player state across frames — a technique used in commercial racing and platformer titles.
- Built an adaptive difficulty engine that adjusts obstacle speed and spawn rate based on real-time player performance, keeping session retention high through dynamic challenge.
- Added a hidden speed-runner shortcut path, demonstrating intentional game-design thinking beyond pure coding.

MNMX Memory Battle: AI Card Game | ishanfernandes.uk/projects 2026

- Built a probabilistic AI opponent that tracks previously revealed cards and weights move decisions using Bayesian-style memory, not a random bot.
- Refactored the entire project from a single-file prototype to a clean multi-file architecture mid-development, applying real-world MVC separation principles.
- Implemented 5 progressive difficulty tiers that tune AI aggression and memory retention, extending average session length.

Course Management System | *Academic Project* 2025

- Designed a multi-entity OOP system (students, courses, grades, attendance, rankings) fully modelled in UML before a single line was written.
- Implemented CSV-based persistence and Matplotlib visualisations, delivering a working data pipeline from file I/O to graphical output.
- Applied 3+ design patterns (e.g. separation of concerns, single responsibility), documented in UML diagrams submitted alongside the codebase.

Portfolio Website | ishanfernandes.uk 2025 - Present

- Built entirely from scratch with zero frameworks, pure HTML, CSS, and modular JavaScript across separate files, demonstrating deliberate architectural choices, not convenience.
- Implemented JS animation system, modal stack, form validation, and full responsive layout without a single library.
- Live production site, publicly accessible and not a Figma mock up or localhost demo.

TECHNICAL SKILLS

Languages	JavaScript (advanced) · Python · Java · HTML5 / CSS3 · SQL
Game Dev	Canvas API · Custom game loops · Collision detection · Ghost replay · Adaptive AI · Godot (learning)
Engineering	OOP · Design Patterns · MVC · UML · Git / GitHub
Web	DOM manipulation · REST APIs · Responsive design · CSS animations · Form validation

Security	Network security · Threat modelling · Encryption fundamentals · Ethical hacking concepts
AI / Tools	Probabilistic AI agents · Symbolic AI · GitHub Copilot · Prompt engineering

EDUCATION

BSc Computer Science | Royal Holloway, University of London 2025 – Present

- Specialising in Software Engineering from Year 2; core modules include Algorithms, OOP, AI, Network Security, and Mathematics for CS.
- Year 2 planned modules: Software Engineering · Symbolic AI · Mathematical Methods for CS.
- All major coursework projects shipped as public GitHub repositories with working demos.

Self-Directed Learning | Ongoing

- Transitioning browser game portfolio to engine-based development: actively building in Godot (GDScript + scene system) and exploring Ren'Py for narrative formats.
- Studying offensive and defensive security independently alongside formal degree modules — including ethical hacking methodology and secure-by-design software principles.